

SOQL & SOSL

Querying and searching Salesforce data – syntax, clauses, and limits at a glance.

Basic SOQL Shape

```
SELECT Id, Name, Industry
FROM Account
WHERE Industry = 'Tech'
ORDER BY Name ASC
LIMIT 100
```

- `SELECT` fields are **explicit** – no `SELECT *`.
- Field & object API names are case-insensitive.
- Query in Apex, REST, `sf data query`, or the Query Editor.

WHERE Operators

OPERATOR	MEANING
<code>=</code> <code>!=</code>	equals / not
<code><</code> <code>></code> <code><=</code> <code>>=</code>	comparison
<code>LIKE</code>	wildcard (<code>%</code> , <code>_</code>)
<code>IN</code> / <code>NOT IN</code>	value set / subquery
<code>INCLUDES</code>	multi-select picklist

```
WHERE Name LIKE 'Acme%'
AND AnnualRevenue > 1000000
```

Bind Variables (Apex)

```
String ind = 'Tech';
List<Account> a =
    [SELECT Id FROM Account
     WHERE Industry = :ind];
```

Always bind variables – never concatenate user input (SOQL injection).

Relationship Queries

Child → Parent (dot notation, up to 5 levels):

```
SELECT Name, Account.Owner.Name
FROM Contact
```

Parent → Child (subquery, use the child relationship name):

```
SELECT Name,
       (SELECT LastName FROM Contacts)
FROM Account
```

- Up to **20** parent → child subqueries per query.

Aggregates & GROUP BY

```
SELECT Industry, COUNT(Id) total
FROM Account
GROUP BY Industry
HAVING COUNT(Id) > 5
```

- Functions: `COUNT`, `SUM`, `AVG`, `MIN`, `MAX`.
- Returns `AggregateResult` in Apex; alias then `get('total')`.

Date Literals

```
WHERE CreatedDate = TODAY
WHERE CloseDate = THIS_MONTH
WHERE CreatedDate = LAST_N_DAYS:30
```

- Use literals, not hard-coded dates.
- `YESTERDAY`, `THIS_WEEK`, `NEXT_QUARTER`, `THIS_FISCAL_YEAR`.

Ordering, Nulls & Limits

```
ORDER BY LastModifiedDate DESC NULLS LAST
LIMIT 200 OFFSET 20
```

- `OFFSET` max **2000**; prefer keyset pagination for large sets.
- `FOR UPDATE` locks rows within the transaction.

SOSL — Full-Text Search

```
FIND {Acme OR "Global Media"}
IN ALL FIELDS
RETURNING Account(Name),
          Contact(FirstName, LastName)
```

- Searches **across objects** at once; SOQL queries one object tree.
- Scopes: `ALL FIELDS`, `NAME FIELDS`, `EMAIL FIELDS`, `PHONE FIELDS`, `SIDEBAR FIELDS`.
- Great when you don't know which field holds the term.

Governor Limits

LIMIT	VALUE
SOQL queries / transaction	100 sync · 200 async
Rows retrieved / transaction	50,000
SOSL queries / transaction	20
SOSL rows returned	2,000

Bulkify: query **outside** loops, filter with `IN :collection`.

Tooling

```
sf data query --query \
  "SELECT Id, Name FROM Account LIMIT 5"

sf data query --query "..." --bulk
```

- Add `--result-format csv|json|human`.
- `USING SCOPE mine` limits to the running user's records.